Solving Radical Equations and Checking Extraneous Roots

Isolating and squaring, checking every candidate for extraneous roots, moving between radical and rational-exponent form, and solving applied radical models

Why checking is part of solving. Squaring both sides is not a reversible step: $a = b$ forces $a^2 = b^2$, but $a^2 = b^2$ only says $a = \pm b$. So the squared equation can have roots the original does not — **extraneous roots**. A radical equation is not finished until every candidate has been substituted into the *original* equation.

Show the substitution check for every problem that asks for one, and state units in applied answers.

1. Solve for x, then check your solution in the original equation.

$$\sqrt{x-3} = 5$$

Answer: ________________

2. Rewrite $16^{3/4}$ as a radical power, then evaluate it without a calculator. (Take the root first, then the power — the numbers stay small.)

Answer: ________________

3. The equation $\sqrt{x+6} = x$ leads to the squared equation $x^2 - x - 6 = 0$.

(a) Solve the squared equation to get two candidates. (b) Substitute each candidate into $\sqrt{x+6} = x$ and record the value of $\sqrt{x+6} - x$. (c) State which candidate is extraneous and which is the solution.

Answer: ________________

4. Solve $\sqrt{3x+1} = x - 1$.

(a) Square both sides and simplify to $x^2 - 5x = 0$; solve it. (b) Check both candidates in the original equation and give the value of $\sqrt{3x+1} - (x-1)$ for the one that fails. (c) Write the solution set.

Answer: ________________

5. Free fall (given: no air resistance, dropped from rest). The time t in seconds for an object to fall h feet is $t = \sqrt{h/16}$.

A stone dropped from a bridge takes $t = 2.5$ seconds to reach the water. How high is the bridge, in feet? Solve the radical equation and check the answer in the model.

Answer: ________________ ft

6. Skid marks (given: dry asphalt, straight skid). A car's speed s in miles per hour is $s = \sqrt{24x}$, where x is the length of the skid mark in feet.

An investigator needs the skid length that corresponds to a speed of 60 miles per hour. Solve the radical equation for x.

Answer: ________________ ft

7. Solve, or explain why no real solution exists:

$$\sqrt{x-2} = -3$$

Squaring gives a candidate. Substitute it into the original equation, report the value of $\sqrt{x-2}+3$ there, and use the range of the principal square root to explain the result.

Answer: ________________

8. Solve $x^{3/2} = 8$ for x.

Write the equation in radical form first, then undo the power and the root. State why this equation has no negative solution.

Answer: ________________

9. Solve $\sqrt{x+7} - \sqrt{x} = 1$.

Isolate one radical before squaring, then isolate the remaining radical and square again. Check the candidate in the original equation.

Answer: ________________

10. The solutions of $\sqrt{x+5} = x-1$ are the x-values where the graphs of $y = \sqrt{x+5}$ and $y = x-1$ meet. The table shows both sides at the two candidates that squaring produces.

x	$\sqrt{x+5}$	$x-1$	equal?
-1	$\sqrt{-1+5} = 2$	-2	
4	$\sqrt{4+5} = 3$	3	

(a) Complete the last column. (b) Give the solution set. (c) Squaring turned the line $y = x-1$ into a parabola that meets the curve twice. Explain, using the table, why the extra intersection is not a solution of the original equation.

Answer: ________________